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CMP49412 – Intelligent Recommendation Systems
Assignment 1
Due: September 28, 2025 by 11:59PM

This assignment will focus on **non-personalized recommendations**.

You will use the provided dataset (*user_ratings.csv*), which contains user ratings for movies.

Question 1:

You have just been hired as a data scientist at a streaming platform. Your manager noticed that the current trending movie list is biased toward movies with only a few ratings.

You are tasked with improving the trending list. You decide to implement a Bayesian Scoring system, which stabilizes rankings by factoring in the global average rating and the number of ratings per movie.

Using the dataset (*user_ratings.csv*):

- a) Compute the Bayesian scores for all movies.
- b) Rank movies based on their Bayesian scores.
- c) Recommend the top 5 trending movies for the platform.

Question 2:

Your company has decided to eliminate the 5-star rating system and switch to a simpler thumbs-up/down voting system. While this makes rating movies easier for users, it also introduces a new challenge:

- Some movies with only a few positive votes may rank too high.
- Other movies with many mixed votes may rank unfairly low.

This change makes your Bayesian scoring system no longer applicable, since it relies on numerical ratings. To maintain a fair trending movie list, you decide to implement Wilson Scoring, which accounts for both the number of votes and confidence intervals to ensure that movies with very few votes don't get ranked too high.

Using the dataset (*user_ratings.csv*):

- a) Convert movie ratings into thumbs-up/down:
 - Thumbs-up = Ratings ≥ 4.0
 - Thumbs-down = Ratings < 4.0
- b) Calculate the Wilson score for each movie.
- c) Rank movies based on their Wilson scores.
- d) Recommend the top 5 trending movies.

Question 3:

Your platform wants to introduce a new “Frequently Watched Together” feature, where users who watched one movie are recommended related movies based on past user behavior.

You have been asked with discovering movies that are frequently watched together.

You decide to use the Apriori Algorithm to identify movies that are frequently watched together.

Using the dataset (*user_ratings.csv*):

- a) Convert the dataset into a transactional format by considering users who have rated movies ≥ 4.0 as “watched”.
- b) Use the Apriori Algorithm to extract frequent itemsets (movies often watched together).
- c) Generate up to 3 association rules, such as “Users who watched Movie A also watched Movie B”.

Submission Requirements:

Submit a **Jupyter Notebook** containing:

1. Your code that implements solutions for each question.
2. The resulting recommendations from each question (Bayesian Scoring, Wilson Scoring, and Association Rule Mining).
3. Brief explanations/comments in the notebook to describe how your code works.